RACE Regular Analysis Through Continuous Exercise

Thermodynamics and Thermochemistry : RACE-01

Introduction; Thermodynamic terms; Types of the system

1. Thermodynamics mainly deals with

(A) interrelation of various forms of energy and their transformation from one form to another.

(B) energy changes in the processes which depend only on initial and final states of the microscopic systems containing a few molecules.

(C) how and at what rate these energy transformations are carried out.

(D) the system in equilibrium state or moving from one equilibrium state to another equilibrium state.

CTF158

The State of the system; State function; Path function

2. Which of the following is not a thermodynamic property of a system?

(A) H (B) P (C) E (D) w

CTF173

3. The volume, V, of a right circular cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height of cylinder. Show that

(a) $dV = \pi r^2 dh + 2\pi rh\, dr$ (b) V is state function

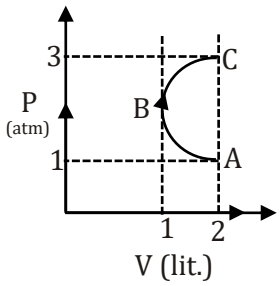
CTF164

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Work Done

4. In the PV-diagram shown in figure,

Find the work done in the process ABC (in *l*-atm)



CTF162

Heat and Heat Capacity (Cm, Cs, Cp, Cv)

5. In an exothermic reaction, heat is evolved, and system loses heat to the surrounding. For such system

(A) q_p will be negative (B) $\Delta_r H$ will be negative

(C) q_p will be positive (D) $\Delta_r H$ will be positive

CTF159

Enthalpy

6. Which of the following statement is/are true for one mole of an ideal gas ?

(A) $\left(\frac{\partial H}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V = R$ (B) $\left(\frac{\partial H}{\partial T}\right)_P > \left(\frac{\partial U}{\partial T}\right)_V$ (C) $\left(\frac{\partial H}{\partial V}\right)_T = 0$ (D) $\left(\frac{\partial H}{\partial P}\right)_T = 0$

CTF161

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First Law of Thermodynamics

7. A system absorbs 20 kJ heat and does 10 kJ of work. The internal energy of the system
(A) increases by 10 kJ (B) decreases by 10 kJ
(C) increases by 30 kJ (D) decreases by 30 kJ

CTF174

8. The volume of a system becomes twice its original volume on the absorption of 300 cal of heat. The work done on the surrounding was found to be 200 cal. What is ΔU for the system?
(A) 500 cal (B) 300 cal (C) 100 cal (D) −500 cal

CTF175

9. A system is provided 50 J of heat and work done on the system is 10 J. The change in internal energy during the process is :-
(A) 40 J (B) 60 J (C) 80 J (D) 50 J

CTF157

Degree of freedom

10. The number of degrees of freedom of molecules in a gas whose molar heat capacity is 29 J/mol-K in the process PT = constant is

CTF176

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ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	AD	D	use Euler theorem	$\left(-\frac{\pi}{2}\right)$	AB	ABCD	A	C
Question	9	10						
Answer	B	3						

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RACE
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Thermodynamics and Thermochemistry : RACE-02

Poisson's ratio

1. For which of the following gases, the difference in specific heats at constant pressure and at constant volume is equal to 0.04545 cal/gm?

(A) N_2 (B) N_2O (C) CO (D) CO_2

CTF177

Isothermal Process

2. Which of the following is the best container for gas during isothermal process?

(A) Glass vessel (B) Wood vessel (C) Copper vessel (D) Thermos flask

CTF178

3. Mark as true or false with appropriate reason -

Assertion	T/F	Reason
1. ΔH is a state function		
2. C_V is independent of T for a perfect (ideal) gas.		
3. $\Delta U = q + w$ for every thermodynamic system at rest in the absence of external field.		
4. A process in which final temperature equals initial temperature must be an isothermal process		
5. U remain constant in every isothermal process in a closed system		
6. q = 0 for every cyclic process.		
7. $\Delta U = 0$ for every cyclic process.		
8. A thermodynamic process is specified by specifying only initial and final state of system.		
9. P-V work is usually negligible for solid and liquid.		
10. If neither heat nor matter can enter or leave a system, that system must be isolated.		

CTF163

4. A balloon containing air at 27°C and 1 bar initially is filled with air further, isothermally and reversibly till the pressure is 4 bar. If the initial diameter of the balloon is 1 m and the pressure at each stage is proportional to diameter of balloon, the magnitude of work done (in 10^7 J) is

CTF179

5. The final volume (in L) of one mole of an ideal gas initially at 27°C and 8.21 atm pressure, if it absorbs 420 cal of heat during a reversible isothermal expansion is (ln 2 = 0.7)

CTF180

Isochoric process

6. Three moles of an ideal gas being initially at a temperature 273 K was isothermally expanded 5.0 times its initial volume and then isochorically heated so that the pressure in the final state became equal to that in the initial state. The total amount of heat transferred to the gas during the process equals 80 kJ. The value of $C_{v,m}$ for this gas (in cal/K-mol) is (ln 5 = 1.6)

CTF181

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Isobaric process

7. A quantity of 70 calories of heat is required to raise the temperature of 2 mole of an ideal gas at constant pressure from 40°C to 50°C. The amount of heat required to raise the temperature of the same gas through the same range at constant volume is

(A) 90 calorie (B) 70 calorie (C) 50 calorie (D) 30 calorie

CTF182

8. One mole of an ideal gas undergoes the following cyclic process.

(i) Isochoric heating from (P_1, V_1, T_1) to double temperature.

(ii) Isobaric expansion to double volume.

(iii) Linear expansion (on P–V curve) to (P_1, $8V_1$).

(iv) Isobaric compression to initial state.

If T_1 = 300 K, then the magnitude of net work done by the gas in the cyclic process is

(A) 2400 cal (B) 1200 cal (C) 4800 cal (D) 3600 cal

CTF183

9. A quantity of 8 g oxygen gas is expanded isothermally at 27°C from 2 dm^3 to 8 dm^3 at a constant external pressure of 4 bar. If the magnitude of work done in this process is used in lifting a body of mass 40 kg, the height (in metre) up to which the body can be lifted is (g = 10 ms^{-2})

CTF184

10. Ten moles of a certain ideal gas at a temperature 300 K was cooled isochorically so that the gas pressure reduced 2.0 times. Then, as a result of the isobaric process, the gas expanded till its temperature got back to the initial value. The total amount of heat absorbed (in Kcal) by the gas in this process is

CTF185

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	BD	C	**1.** False **2.** False **3.** False **4.** False **5.** False **6.** False **7.** True **8.** False **9.** True **10.** False	1	6	5	D	A	6	3

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Regular Analysis Through Continuous Exercise

Thermodynamics and Thermochemistry : RACE-03

Adiabatic process

1. Which of the following is true for reversible adiabatic process involving an ideal gas?
(A) Gas with higher γ has high magnitude of slope in a P (y-axis) vs. T (x-axis) curve.
(B) Gas with higher γ has high magnitude of slope in a V (y-axis) vs. T (x-axis) curve.
(C) Gas with higher γ has high magnitude of slope in a P (y-axis) vs. V (x-axis) curve.
(D) Gas with higher γ has low magnitude of slope in a P (y-axis) vs. T (x-axis) curve.

CTF186

Comparison between Isothermal and Adiabatic process

2. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its absolute temperature. The Poisson's ratio of gas is
(A) 3/2 (B) 7/2 (C) 5/3 (D) 9/7

CTF187

3. In the reversible adiabatic expansion of an ideal monoatomic gas, the final volume is 8 times the initial volume. The ratio of final temperature to initial temperature is
(A) 8 : 1 (B) 1 : 4 (C) 1 : 2 (D) 4 : 1

CTF188

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4. Two moles of an ideal gas (γ= 1.4) was allowed to expand reversibly and adiabatically from 1 L, 527°C to 32 L. The molar enthalpy change of the gas is
(A) −4200R (B) −2100R (C) −1500R (D) −3000R

CTF189

5. Equal moles of He, H_2, CO_2 and SO_3 gases are expanded adiabatically and reversibly from the same initial state to the same final volume. The magnitude of work is maximum for (Assume ideal behaviour of gases and all the degree of freedoms are active.)
(A) He (B) H_2 (C) CO_2 (D) SO_3

CTF190

6. Two moles of an ideal gas (Cv,m = 1.5R) is subjected to the following changes in states.

$$\text{A (500 K, 5 bar)} \xrightarrow[\text{Expansion}]{\text{Reversible isothermal}} \text{B}$$

$$\xrightarrow{\text{Isochoric cooling}} \text{C}(250\text{K}, 1\text{bar}) \xrightarrow[\text{Compression}]{\text{Single stage a adibatic}} \text{D (3 bar)}$$

The correct statement(s) is/are
(A) The pressure at B is 2.0 bar.
(B) The temperature at D is 450 K.
(C) ΔH_{CD} = 1000R
(D) ΔU_{BC} = 375R

CTF191

Polytropic process

7. An ideal gas undergoes a process in which its pressure and volume are related as PV^n= constant, where n is a constant. The molar heat capacity for the gas in this process will be zero if
(A) $n = \gamma$ (B) $n = \gamma - 1$ (C) $n = \gamma + 1$ (D) $n = 1 - \gamma$

CTF192

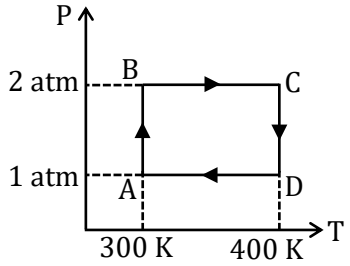
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8. An ideal monoatomic gas undergoes a reversible process, where $\frac{P}{V}$ = constant, from (2 bar, 273 K) to 4 bar. The value of $\frac{\Delta U}{w}$ for this process is

(A) +3.0 (B) −3.0 (C) −1.5 (D) +1.5

CTF193

9. Two moles of helium gas undergoes a cyclic process as shown in the figure. Assuming ideal behaviour of gas, the magnitude of net work done by the gas in this cyclic process is



(A) 0 (B) 100R ln2 (C) 100R ln4 (D) 200R ln4

CTF194

10. An ideal gas is taken reversibly from state A (P, V) to the state B (0.5 P, 2 V) along a straight line in PV diagram. Which of the following statement(s) is/are correct regarding the process?

(A) The work done by gas in the process A to B exceeds the work that would be done by it if the same change in state were performed isothermally.

(B) In the T–V diagram, the path AB becomes a part of parabola.

(C) In the P–T diagram, the path AB becomes a part of hyperbola.

(D) On going from A to B, the temperature of the gas first increases to a maximum value and then decreases.

CTF195

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	CD	A	B	B	D	ABC	A	B	C	ABD

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RACE
Regular Analysis Through Continuous Exercise

Thermodynamics and Thermochemistry : RACE-04

Introduction; Exothermic and Endothermic Reaction

1. The word standard in molar enthalpy change implies

(A) temperature 298 K and pressure 1 bar.

(B) any temperature and pressure 1 atm.

(C) any temperature and pressure 1 bar.

(D) any temperature and pressure.

CTC174

Relationship Between ΔH and ΔU; ΔH=ΔU+ΔngRT for Chemical Reaction

2. How much heat (in kJ) should be supplied to a rigid conducting vessel of volume V litres, so that 0.5 mole of C is formed in the reaction.

$A_{(g)} + B_{(g)} \rightarrow C_{(g)} : \Delta H = +7.6$ kJ

Take R = 8 J/K mole , T = 300 K :

CTC149

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Enthalpy of Combustion

3. Enthalpies of combustion of CH_4, C_2H_4 and C_2H_6 are −890, −1411 and −1560kJ/ mole, respectively. Which has the highest fuel value (heat produced per gram of the fuel)?

(A) CH_4 (B) C_2H_6

(C) C_2H_4 (D) All are same

CTC175

4. Calculate the heat of combustion (kJ) of propane, C_3H_8 using the listed standard enthalpy of reaction data: $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(g)$

$3\ C(s) + 4\ H_2(g) \rightarrow C_3H_8(g)$ ΔH^o/kJ = – 103.8

$C(s) + O_2(g) \rightarrow CO_2(g)$ ΔH^o/kJ = – 393.5

$H_2(g) + 1/2\ O_2(g) \rightarrow H_2O(g)$ ΔH^o/kJ = – 241.8

(A) – 2043.9 (B) – 1532.9 (C) – 1021.9 (D) –739.1

CTC144

5. A geyser, operating on LPG (liquefied petroleum gas) heats water flowing at the rate of 3.0 litres per minute, from 27°C to 77°C. If the heat of combustion of LPG is 40,000 J/g, then how much fuel (in g) is consumed per minute? (Specific heat capacity of water is 4200 J/kg-K)

(A) 15.25 (B) 15.50 (C) 15.75 (D) 16.00

CTC176

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6. In certain areas where coal is cheap, artificial gas is produced for house hold use by the 'water gas' reaction.

$C(s) + H_2O(g) \longrightarrow H_2(g) + CO(g)$

Assume that coke is 100% carbon. Given ΔH_C(kcal/mol) at 25°C: $H_2(g) = -68.0$; $CO(g) = -68.0$; $C(s) = -94.0$. Select the correct option(s)from the following.

(A) The maximum heat obtainable at 25°C from the combustion of 1.2 kg of coke is 9400 kcal.

(B) The maximum heat obtainable at 25°C from burning water gas produced from 1.2 kg of coke is 13,600kcal.

(C) The maximum heat obtainable at 25°C from burning 1.2 kg of water gas is 5440 kcal.

(D) The maximum heat obtainable at 25°C from burning water gas or burning the same mass of coke will be same.

CTC177

7. In order to get maximum calorific output, a burner should have an optimum fuel to oxygen ratio which corresponds to 3 times as much oxygen as required theoretically for complete combustion of the fuel. A burner which has been adjusted for methane as fuel (with 'X' L/hr of methane and '6X' L/hr of oxygen) is to be adjusted for butane. In order to get same calorific output, what should be the supply of oxygen (in L/hr) with 1 L/ hr supply of butane? Assume that losses due to incomplete combustion, etc., are the same for both fuels and that the gases behave ideally. Enthalpies of combustion; methane = −800 kJ/mol; butane = −3120 kJ/mol.

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CTC178

Enthalpy of Formation

8. The fat, glyceryl trioleate, is metabolized via the following reaction. Given the enthalpies of formation, calculate the energy (kJ) liberated when 1.00 g of this fat reacts.

(Atomic weights: C = 12, H = 1, O = 16).

$C_{57}H_{104}O_6(s) + 80\ O_2(g) \rightarrow 57\ CO_2(g) + 52\ H_2O(l)$

$\Delta H^\circ\ C_{57}H_{104}O_6 = -70870$ kJ/mole

$\Delta H^\circ\ H_2O(l) = -285.8$ kJ/mole

$\Delta H^\circ\ CO_2(g) = -393.5$ kJ/mole

(A) 40.42 (B) 33.40

(C) 37.98 (D) 42.62

CTC142

For More Join: @IITJEE_Advanced

9. Determine ΔH° of the following reaction using the listed heats of formation :

$4\ HNO_3(l) + P_4O_{10}(s) \rightarrow 2\ N_2O_5(s) + 4\ HPO_3(s)$

$\Delta H^{\circ}_f\ HNO_3(l) = -174.1$ kJ/mole

$\Delta H^{\circ}_f\ N_2O_5(s) = -43.1$ kJ/mole

$\Delta H^{\circ}_f\ P_4O_{10}(s) = -2984.0$ kJ/mole

$\Delta H^{\circ}_f\ HPO_3(s) = -948.5$ kJ/mole

(A) –176.3 (B) – 199.8 (C) +276.2 (D) – 242.4

CTC143

10. Study the following thermochemical equations.

$A \longrightarrow B;\ \Delta H = +100$ kcal

$B \longrightarrow C;\ \Delta H = -80$ kcal

The correct order of enthalpies of formation of A, B and C is

(A) A < B < C (B) A < C < B (C) C < A < B (D) B < C < A

CTC179

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	5	A	A	C	ABC	5	C	B	B

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RACE
Regular Analysis Through Continuous Exercise

Thermodynamics and Thermochemistry : RACE-05

Enthalpy of Formation

1. Formation of ozone from oxygen is an endothermic process. In the upper atmosphere, ultraviolet is the source of energy that drives the reaction. Assuming that both the reactions and the products of the reaction are in standard states, the standard enthalpy of formation of ozone from the following information.

$3O_2(g) \longrightarrow 2O_3(g)$, $\Delta H° = 286$ kJ, is

(A) +143 kJ/mol (B) −143 kJ/mol (C) +286 kJ/mol (D) −286 kJ/mol

CTC180

2. The value of $\Delta_f H°$ of $U_3O_8(s)$ is −853.5kJ mol^{-1}. $\Delta H°$ for the reaction

$3UO_2(s) + O_2(g) \longrightarrow U_3O_8(s)$

is −76.00 kJ. The value of $\Delta_f H°$ of $UO_2(s)$ is

(A) −259.17 kJ/mol (B) −310.17 kJ/mol (C) +259.17 kJ/mol (D) 930.51 kJ/mol

CTC181

3. The standard enthalpies of formation of $H_2O(l)$, $Li^+(aq)$ and $OH^-(aq)$ are −285.8, −278.5 and −228.9 kJ/mol, respectively. The standard enthalpy change for the below reaction is

$2Li(s) + 2H_2O(l) \longrightarrow 2Li+(aq) + 2OH^-(aq) + H_2(g)$

(A) +443.2 kJ (B) −443.2 kJ (C) −221.6 kJ (D) +221.6 kJ

CTC182

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4. Benzene burns in oxygen according to the following reactions.

$$C_6H_6(l)+\frac{15}{2}O_2(g)\longrightarrow 3H_2O(l)+6CO_2(g)$$

If the standard enthalpies of formation of $C_6H_6(l)$, $H_2O(l)$ and $CO_2(g)$ are 11.7, −68.1 and −94 kcal/mole, respectively, the amount of heat that will liberate by burning 780 g of benzene is

(A) 7800 kcal (B) 780 kcal (C) 78 kcal (D) 608.4 kcal

CTC183

5. For which of the following substance(s), the standard enthalpy of formation is/are zero?

(A) C (graphite) (B) $O_3(g)$ (C) $I_2(g)$ (D) $Br_2(l)$

CTC184

Measurement of ΔU and ΔH; Calorimetry; Bomb Calorimeter

6. When 0.18 g of glucose was burnt in a bomb calorimeter, the temperature rose by 4 K. The heat capacity of the calorimeter system is ($\Delta_C H°$=−2.8× 10^6J/ mol glucose)

(A) 700 J/K (B) 700 kJ/K (C) 126 J/K (D) 7.0 kJ/K

CTC185

Bond Enthalpy

7. Use the given bond enthalpy data to estimate the ΔH^o (kJ) for the following reaction.

(C – H = 414 kJ, H – Cl = 431 kJ, Cl – Cl = 243 kJ, C – Cl = 331 kJ).

$CH_4(g) + 4\ Cl_2(g) \rightarrow CCl_4(g) + 4\ HCl(g)$

(A) 620 (B) 330 (C) 420 (D) 105

CTC147

For More Join: @IITJEE_Advanced

Enthalpy of Hydrogenation

8. The enthalpy of combustion at 25°C of $H_2(g)$, cyclohexane(l) and cyclohexene(l) are −241, −3920 and −3800 kJ/ mol, respectively. The enthalpy of hydrogenation of cyclohexene(l) is

(A) −121 kJ/mol (B) +121 kJ/mol (C) −242 kJ/mol (D) +242 kJ/mol

CTC186

Enthalpy of Neutralisation

9. A solution of 500 ml of 2 M–KOH is added to 500 ml of 2 M–HCl and the mixture is well shaken. The rise in temperature T_1 is noted. The experiment is again performed using 250 ml of KOH solution and 500 ml of HCl solution and rise in temperature T_2 is again noted. Which of the following is correct?

(A) $T_1 = T_2$ (B) T_1 is 2 times as larger as T_2.

(C) T_2 is twice larger as T_1. (D) T_1 is 1.5 times as larger as T_2.

CTC187

Resonance Energy

10. Consider the following isomerization process.

$CH_2=CH-CH_2-CH=CH_2(g) \longrightarrow CH_2=CH-CH=CH-CH_3(g)$

Which of the following statement(s) is/are true regarding this process?

(A) The process is exothermic.

(B) Enthalpy change of reaction = Resonance enthalpy of product.

(C) The magnitude of enthalpy of combustion of product is less than that of reactant.

(D) The magnitude of enthalpy of hydrogenation of product is greater than that of reactant.

CTC188

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	A	B	A	AD	A	C	A	D	ABC

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RACE
Regular Analysis Through Continuous Exercise

Thermodynamics and Thermochemistry : RACE-06

Entropy and Spontaneity; Physical Significance of Entropy

1. **Assertion (A) :** A liquid crystallises into a solid and is accompanied by decrease in entropy.
Reason (R) : In crystals, molecules organise in an ordered manner.
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not the correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

CTS166

2. The spontaneity means, having the potential to proceed without the assistance of external agency. The processes which occur spontaneously are
(A) flow of heat from colder to warmer body.
(B) gas in a container contracting into one corner.
(C) gas expanding to fill the available volume.
(D) burning carbon in oxygen to give carbon dioxide.

CTS200

3. For which of the following process ΔS°_{sys} is positive?
(A) The dissolution of ammonium nitrate in water.
(B) The expansion of a sample of chlorine into a larger volume.
(C) The decomposition of mercuric oxide.
(D) The rusting of iron.

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CTS186

Second Law of Thermodynamics

4. A system undergoes a process in which the entropy change is +5.51 JK^{-1}. During the process, 1.50 kJ of heat is added to the system at 300 K. The correct information regarding the process is
(A) the process thermodynamically reversible.
(B) the process is thermodynamically irreversible.
(C) the process may or may not be thermodynamically reversible.
(D) the process must be isobaric.

CTS187

5. List equation/law (in Column I) with statement (in Column II) :

Column-I		Column-II	
(A)	Arrhenius equation	(p)	Variation of enthalpy of a reaction with temperature
(B)	Kirchhoff equation	(q)	Variation of rate constant with temperature
(C)	Second law of thermodynamics	(r)	Entropy of an isolated system tends to increase and reach a maximum value
(D)	Hess's law of constant heat summation	(s)	Enthalpy change in a reaction is always constant and independent of the manner in which the

CTS171

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6. Which of the following statement(s) is/are incorrect?
(A) Entropy is a measure of unavailable energy.
(B) Decrease in free energy of the system represents the unavailable energy of the system.
(C) Complete conversion of heat into work is not possible in a reversible isothermal expansion of ideal gas.
(D) Complete conversion of heat into work is not possible in an irreversible isothermal expansion of ideal gas.

CTS188

Entropy Change for an Ideal gas; Isothermal; Isochoric; Isobaric; Adiabatic Process

7. When one mole of an ideal gas is compressed to half of its initial volume and simultaneously heated to twice its temperature, the change in entropy is
(A) $C_{V,m}$ ln2 (B) $C_{P,m}$ ln2 (C) R ln2 (D) ($C_{V,m}$ − R) ln2

CTS189

8. The standard molar entropy of an ideal gas $\left(\gamma = \frac{4}{3}\right)$ is 2.5 cal/K mol at 25°C and 1 bar. The standard molar entropy of the gas at 323°C and 1 bar is (ln 2 = 0.7)
(A) undefined (B) 4R ln2 (C) 5.6 cal/K-mol (D) 8.1 cal/K-mol

CTS190

9. Molar heat capacity of CD_2O (deuterated form of formaldehyde) vapour at constant pressure is 14 cal/K-mol. The entropy change associated with the cooling of 3.2 g of CD_2O vapour from 1000 K to 900 K at constant pressure is (assume ideal gas behaviour for CD_2O) [ln 0.9 = −0.1]
(A) +0.14 cal/K (B) – 0.14 cal/K (C) −1.4 cal/K (D) +1.4 cal/K

CTS191

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Entropy Change for Phase Transformation

10. The entropy change can be calculated by using the expression $\Delta S = \frac{q_{rev}}{T}$. When water freezes in a glass beaker, choose the correct statement amongst the following :
(A) ΔS (system) decreases but ΔS (surroundings) remains the same.
(B) ΔS (system) increases but ΔS (surroundings) decreases.
(C) ΔS (system) decreases but ΔS (surroundings) increases.
(D) ΔS (system) decreases and ΔS (surroundings) also decreases.

CTS163

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	A	CD	ABC	B	A → q, B → p, C → r, D → s	BCD	D
Question	8	9	10				
Answer	D	B	C				

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RACE
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Thermodynamics and Thermochemistry : RACE-07

Entropy Change for Phase Transformation

1. What is the entropy change when 3.6 g of liquid water is completely converted into vapours at 373 K? The molar heat of vaporization is 40.85 kJ/mol.
(A) 219.0 J/K (B) 2.190 J/K (C) 21.90 J/K (D) 109.5 J/K

CTS192

2. One mole of ice is converted into water at 273 K and 1 atm. The entropies of H_2O (s) and H_2O (l) are 38.0 and 58.0 J/K-mol, respectively. The enthalpy change for the conversion is
(A) 5460 kJ/mol (B) 5460 J/mol (C) −5460 J/mol (D) 20 J/mol

CTS193

Entropy Change for Chemical Reaction

3. Given the following entropy values (in J/K-mol) at 298 K and 1 atm $H_2(g)$ = 130.6, $Cl_2(g)$ = 223.0 and HCl(g) = 186.7. The entropy change (in J/K-mol) for the reaction $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$ is
(A) +540.3 (B) +727.0 (C) −166.9 (D) +19.8

CTS194

Gibb's Free Energy and Spontaneity; Calculation of Change in G (ΔG)

4. Which of the following are correct at 298K
(A) $\Delta G°_F (H_{2(g)}) = 0$ (B) $\Delta_F G° (H_2O, l) = 0$
(C) $\Delta_F G° (D_2, g) = 0$ (D) $\Delta_F G° (H^+, aq.) = 0$

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CTS168

5. Match the following parameters with description for spontaneity :

Δ (Parameters)				Description
	$\Delta_r H°$	$\Delta_r S°$	$\Delta_r G°$	
(A)	+	−	+	(i) Non-spontaneous at high temperature.
(B)	−	−	+ at high T	(ii) Spontaneous at all temperatures
(C)	−	+	−	(iii) Non-spontaneous at all temperatures

CTS169

6. The ΔG in the process of melting of ice at −15°C and 1 atm, is
(A) less than zero. (B) greater than zero. (C) equal to zero. (D) None of these

CTS195

7. The values of ΔG are very important in metallurgy. The ΔG values for the following reactions at 1000 K are given as follows.
$S_2(s) + 2O_2(g) \rightarrow 2SO_2(g)$; ΔG = −544 kJ
$2Zn(s) + S_2(s) \rightarrow 2ZnS(s)$; ΔG = −293 kJ
$2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$; ΔG = −480 kJ
The ΔG for the reaction $2ZnS(s) + 3O_2(g) \rightarrow 2ZnO(s) + 2SO_2(g)$ will be
(A) −357 kJ (B) −731 kJ (C) −773 kJ (D) −229 kJ

CTS196

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8. How much energy (in kJ) is available for sustaining muscular and nervous activity from the combustion of 1.0 mole of glucose molecules under standard conditions at 37°C (body temperature)? The standard entropy and enthalpy of reaction are +200 J/K-mol and – 2808 kJ/ mol, respectively.

CTS197

Carnot Cycle

9. An ideal gas undergoes through following cyclic process.

1–2 : Reversible adiabatic compression from $P_1V_1T_1$ to $P_2V_2T_2$

2–3 : Reversible isochoric heating from $P_2V_2T_2$ to $P_3V_3T_3$

3–4 : Reversible adiabatic expansion from $P_3V_3T_3$ to $P_4V_4T_4$

4–1 : Reversible isochoric cooling from $P_4V_4T_4$ to $P_1V_1T_1$

Efficiency of the cycles is :-

(A) $1-\left(\frac{T_4-T_1}{T_3-T_2}\right)$ (B) $1-\frac{T_1}{T_2}$ (C) $1-\left(\frac{V_2}{V_1}\right)^{\gamma-1}$ (D) $1-\frac{|Q_{4-1}|}{|Q_{2-3}|}$

CTS167

10. In which case will the efficiency of a Carnot cycle be higher, when the hot body temperature is increased by ΔT or when the cold body temperature is decreased by the same magnitude?

(A) Same in both.

(B) On decreasing the temperature of cold body.

(C) On increasing the temperature of hot body.

(D) Depends on the initial temperature of cold and hot bodies.

CTS198

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11. What is the efficiency of a cycle consisting of two isochoric and two adiabatic lines, if the volume of the ideal gas changes 10 times within the cycle? The working substance is nitrogen $[(10)^{0.4} = 2.5]$.

(A) 40% (B) 25% (C) 60% (D) 75%

CTS199

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	C	B	D	ACD	A → (i, iii), B → (i), C → (ii)	B	B
Question	8	9	10	11			
Answer	2870	ABCD	B	C			

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